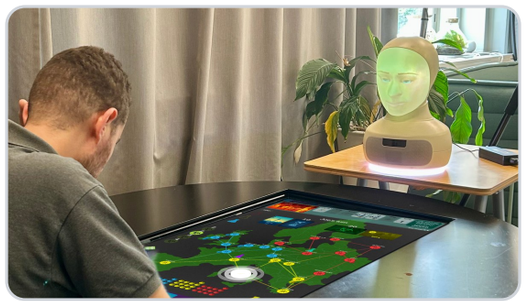
1 Testbed



state ↓ ↑ moves

rule-based solver

HRI corpus
70 sessions · 36 participants
speech · video · gaze · game logs

2 Partner signals

training time only · predicted (RQ1)

a Self-reported

enjoyment ●●●●○
rated once, after each game

b Measured

brain activity
fMRI, only in the scanner

3 Model adaptation *current/future work*

trained

full-duplex

h_t

partner speech
partner gaze
task state

LLM

text head
gaze head
action head

bias

robot speech
robot gaze
robot action

stage 1 · ground
$\pi \cdot v$ self-play
conversational gaze
general · *all heads*
fine-tuned on abundant data

stage 2a · imitate
intervention head
80 ms
general

stage 2b · RL
reward from
predicted partner
state ②
partner-specific